

Your Starter Pack to Agentic Automation [Tamil Edition]

Autonomous Agent — Use Case & Problem Statement

AI News Headline Summary Agent | API-Workflow Domain (Corporate Communications)

Detail	Description
Audience	Engineering & Arts students (challenge exercise)
Agent Type	Autonomous Agent (no human-in-the-loop)
Platform	UiPath Autopilot / Agent Builder
Difficulty	Moderate — introduces live external news aggregation API integration

1. Problem Statement

Corporate communications and leadership teams frequently need a quick digest of the latest news on a specific topic or industry — for example, checking today's headlines about a competitor, an industry regulation change, or a sector the company operates in, before a morning briefing or board meeting. Today, staff manually browse multiple news websites, skim several articles, and compile a summary by email — a time-consuming task that must be repeated daily and does not scale when several topics need to be tracked at once.

Design and build an Autonomous UiPath Agent, the AI News Headline Summary Agent, that reads an incoming email requesting a news summary on a topic, calls a live News API workflow to fetch recent headlines, summarizes the most relevant ones, and independently replies with a concise digest — without any human approval step.

2. System Prompt

You are an AI News Headline Summary Agent for the Corporate Communications team. Your goal is to analyze news summary request emails and provide an accurate, up-to-date headline digest using a live news API.

Follow this reasoning process:

1. Read and understand the email content.
2. Classify it into one of the following:
 - Topic News Summary Request
 - Competitor/Industry Watch Request
 - Other
3. If it is Topic News Summary Request or Competitor/Industry Watch Request:
 - Extract the topic, keyword, or company name and the desired time window (e.g., today, last 7 days)
 - Call the News API workflow with the topic and time window
 - Select up to 5 of the most relevant, recent headlines
 - Generate a professional reply with each headline, a one-line summary, the source, and the publish date
4. If it is Other:
 - Skip processing

Rules:

- Think step-by-step before answering
- Be concise and structured
- Do not assume missing details; if the topic is unclear, ask the requester to clarify instead of guessing
- Do not invent headlines or sources – always use the API response
- Return output in structured format

3. User Prompt

```
Analyze the following news summary request email and provide the latest relevant headlines using the live news API:
{{news_email_text}}
```

4. Input Schema

```
{
  "news_email_text": "string"
}
```

5. Output Schema

```
{
  "category": "string",
  "topic": "string",
  "time_window": "string",
  "headlines": [
    {
      "title": "string",
      "summary": "string",
      "source": "string",
      "published_date": "string"
    }
  ],
  "action": "string"
}
```

6. Tool: API Workflow & Publish

Create the following process inputs for the API workflow (HTTP Request activity) the agent will invoke:

- topic (text)
- time_window (text) — e.g., "today", "7d", "30d"

Configure the workflow to call a live news API endpoint (e.g., an HTTPS GET request to a news-aggregation provider) and parse the JSON response for article titles, summaries, sources, and publish dates.

7. Add Tool: Map Inputs

Agent Field	API Request Parameter
topic	search_query
time_window	date_range

8. Updated System Prompt — Tool Invocation Logic

```
If the email is classified as Topic News Summary Request or
Competitor/Industry Watch Request:
- Call the get_news_headlines API workflow with topic and
  time_window.
- Parse the API response for article 'title', 'description',
  'source', and 'published_at' fields.
- Rank and select up to 5 of the most relevant, recent articles.
- Use the generate_reply GenAI activity to draft a professional
  reply email listing each headline with a one-line summary,
  source, and publish date – fully autonomously, with no manual
  approval required.

If the email is classified as Other:
- Take no further action.
```

9. Tools & Activities Summary

Tool / Activity	Type
get_news_headlines	API Workflow (HTTP Request to a live news aggregation API)
generate_reply	GenAI Activity (LLM prompt-based text generation)
classify_request	GenAI Activity (embedded in Agent reasoning)

Note: This use case is intentionally fully autonomous — the agent classifies the request, calls the live news API, ranks and summarizes headlines, and composes the reply without an escalation or human-review step, demonstrating end-to-end autonomous decision-making for the workshop challenge.

10. Expected Learning Outcome

- Design a system prompt that integrates a live external news API into agent reasoning
- Configure an API Workflow (HTTP Request activity) with mapped input parameters
- Parse a JSON API response containing a list of articles and rank them by relevance
- Define structured input/output schemas for a list-based API-integrated autonomous agent
- Publish and test a fully autonomous, API-driven agent in UiPath Agent Builder